Baryon Magnetic Moments in a Two-Zone Kelvin–Voigt Vortex Model

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Abstract

This work develops a self-consistent extension of the Kelvin–Voigt (K–V) mechanical analogy for Maxwell's equations derived in [6] to the description of the magnetic moments of baryons. Each nucleon is modelled as a single Hill-type vortex comprising a compact inner zone (core, radius r_c) and an extended outer zone (mantle, outer radius r_m), whose parameters are fixed by electron-scattering data on the charge distribution. Within the K–V framework, the electric charge of a vortex ring is the mass-normalised circulation invariant $Q = \rho_0 r_0 \Gamma$; the magnetic moment is derived from the peripheral velocity of the vortex core. The equal-charge condition yields a parameter-free relation between the magnetic moment ratio and the geometric size ratio of the two zones, in quantitative agreement with experiment. The model is extended to the hyperon sector: a falsifiable prediction is formulated for the Λ magnetic form factor, the Σ triplet is shown to require independent charge-distribution data, and the Ξ doublet is found to be incompatible with the equal-charge ansatz under the K–V sign convention; in contrast to the Σ case, where the failure is quantitative, the Ξ system admits no physical solution at all. Concrete proposals for experimental verification are developed. The approach is phenomenological and does not claim to be fundamental, but possesses significant potential for educational applications.

1 Introduction

Experimental studies of electron scattering on nucleons [1, 2], including the pioneering work of Hofstadter and the review by Schopper [3], have revealed a complex internal structure of the proton and neutron, indicating the presence of a compact core ("kernel") and an extended envelope ("mantle") [4]. The standard interpretation of these data relies on the quark model.

In this work, we propose an alternative, phenomenological approach, building on the Kelvin–Voigt mechanical analogy for Maxwell's equations rigorously derived in [6]. In that framework a dilute assembly of compact vortex rings in an incompressible K–V viscoelastic medium obeys equations formally identical to Maxwell's equations; the electric charge is identified with the mass-normalised circulation invariant $Q = \rho_0 r_0 \Gamma$. A key development presented here is the self-consistent extension of this framework from the description of macroscopic electromagnetic fields to the modelling of intrinsic particle properties—in particular, the magnetic moments of nucleons. The aim of the work is to construct an intuitively understandable and mathematically self-consistent model that can serve as a bridge between classical and quantum concepts in the educational process.

2 Theoretical Basis of the Model

2.1 Mathematical Framework

The basis of the model is the K–V mechanical analogy of [6], in which the quantum vacuum is treated as a Kelvin–Voigt viscoelastic medium supporting compact, quantised vortex rings [7]. The two central results of [6] used here are:

- **Electric charge** of a vortex ring is the mass-normalised circulation invariant

$$Q = \rho_0 \, r_0 \, \Gamma, \tag{1}$$

where ρ_0 is the medium density, r_0 is the vortex core radius, and Γ is the circulation, conserved by Kelvin's theorem. For a compact toroidal vortex whose core rotates with angular velocity ω, the circulation around the core is $\Gamma = 2\pi r_0^2 \, \omega$, so that

$$Q = 2\pi \rho_0 \, r_0^3 \, \omega = \tfrac{3}{2} \rho_0 \, V \, \omega, \tag{2}$$

where $V \sim \frac{4}{3}\pi r_0^3$ is the vortex core volume and the numerical prefactor depends on the exact vortex geometry. Equation (2) shows that the K–V charge is proportional to the product of medium density, vortex volume, and angular velocity—the dimensional structure used throughout this paper.

- **Magnetic moment.** The magnetic moment of a current loop of area A carrying current I is $\mu = IA$. In the K–V analogy, the charge of a vortex ring is carried by fluid parcels in the core that rotate with angular velocity ω. For a compact torus ($\rho_{\text{ring}} \sim r_0$), each parcel completes an orbit of radius ρ_{ring} at this same frequency $\omega/(2\pi)$, so the "current" is $I = Q\,\omega/(2\pi)$ and the enclosed area is $A = \pi \rho_{\text{ring}}^2$. For $\rho_{\text{ring}} \sim r_0$:

$$\mu = \rho_0 \, V \, (\omega r_0)^2, \tag{3}$$

where ωr_0 is the peripheral velocity at the vortex core boundary. This expression highlights the unified mechanism generating charge and magnetic moment through vortex rotation: the charge scales with ω (eq. (2)), the magnetic moment with $\omega^2 r_0^2$ (eq. (3)).

Note on sign convention. Equation (3) gives the *magnitude* of the magnetic moment. The sign of μ along the nucleon spin axis is determined by the handedness of the circulation: a vortex with circulation $\Gamma > 0$ (counter-clockwise when viewed from the $+z$ direction) contributes a positive μ_z, while reversing the circulation ($\Gamma \to -\Gamma$) reverses the sign of μ_z. The signed formula is therefore

$$\mu = \text{sign}(\Gamma) \cdot \rho_0 \, V \, (\omega r_0)^2, \tag{4}$$

where $\omega = |\Gamma|/(2\pi r_0^2) > 0$ by definition. This sign convention underlies the additive rules $\mu_p = \mu_c + \mu_m$ (both zones co-rotating) and $\mu_n = -\mu_c + \mu_m$ (core zone reversed) used throughout Section 3.1.

Note on approximation order. Equations (2) and (3) treat each zone as a uniformly rotating body characterised by a single representative angular velocity (ω_c or ω_m) and the corresponding outer radius. For a realistic vortex with a smooth circulation profile, corrections to this uniform-rotation approximation are of relative order $r_c/r_m \approx 0.19$. The uncertainty of the geometric ratio r_m/r_c from Hofstadter-era

charge-distribution data is $\delta(r_m/r_c)/(r_m/r_c) \approx 23\%$ (eq. (12)), which exceeds the approximation error. The leading-order uniform-rotation estimate is therefore consistent with the precision of the available experimental input, and no higher-order correction is warranted.

Note on dimensions. In K–V model units, formula (3) has dimensions of energy:

$$\mu_{\mathrm{KV}} = I_{\mathrm{KV}} \cdot A = \underbrace{[\mathrm{kg\,s^{-2}}]}_{I_{\mathrm{KV}}} \cdot \underbrace{[\mathrm{m^2}]}_{A} = \mathrm{kg\,m^2\,s^{-2}} = \mathrm{J},$$

where the K–V current unit $\mathrm{kg\,s^{-2}}$ is taken from the correspondence table of [6] (row "Current I"). Applying the conversion $I_{\mathrm{SI}} = R\, I_{\mathrm{KV}}$ (where $R = \tau_{\mathrm{KV}} = \eta_0/G$, the Kelvin–Voigt relaxation time, also from that table) gives

$$\mu_{\mathrm{SI}} = I_{\mathrm{SI}} \cdot A = R\, I_{\mathrm{KV}} \cdot A = R\, \mu_{\mathrm{KV}}, \quad [R\, \mu_{\mathrm{KV}}] = \mathrm{s} \cdot \mathrm{J} = \mathrm{kg\,m^2\,s^{-1}} \to \mathrm{A\,m^2},$$

where the last step follows from the SI relation $1\,\mathrm{A} = 1\,\mathrm{kg\,m^2\,s^{-3}\,V^{-1}}$ within the K–V electromagnetic correspondence established in [6]. Since R is the same constant for all zones of the same nucleon, it cancels identically in all dimensionless ratios. Equation (3) is therefore used exclusively to derive ratios such as μ_c/μ_m; absolute values in nuclear magnetons μ_N are taken directly from experiment via the system (7).

2.2 Key Assumptions of the Model

The two-zone structure adopted here is directly motivated by the K–V framework itself. The Hill spherical vortex—the prototype solution employed in [6]—comprises two hydrodynamically distinct regions: an inner *core* of non-zero vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u} \neq 0$ (the rotational domain, characteristic radius r_c), and an outer irrotational region ($\boldsymbol{\omega} = 0$) that nonetheless carries circulation and stores K–V elastic energy (the mantle, outer radius r_m). The nucleon is modelled as a *single* such vortex; its two zones map directly onto the experimentally observed two-scale charge distribution.

1. The nucleon is described by a single Hill-type vortex. Its inner zone (core, radius $r_c \sim 0.25\,\mathrm{fm}$) and outer zone (mantle, outer radius $r_m \sim 1.3\,\mathrm{fm}$) carry the charge and magnetic-moment contributions Q_c, μ_c and Q_m, μ_m respectively. This two-scale structure is clearly visible in the charge distribution data obtained from electron scattering experiments (see Fig. 2).

2. The contributions from each zone are computed from eqs. (2) and (4) as leading-order estimates, treating each zone as a uniformly rotating body with representative angular velocity ω_c or ω_m. The approximation is justified to the precision of the available data (see the approximation note in Sec. 2.1).

3. The proton–neutron transition corresponds to a reversal of the core-zone rotation ($\omega_c \to -\omega_c$, equivalently $\Gamma_c \to -\Gamma_c$) while the mantle zone remains unchanged.

4. The geometric parameters r_c and r_m are fixed by experimental charge-distribution data [4, 5], which ensures the self-consistency of the model.

Note on stability. The Hill spherical vortex is a known steady solution of the incompressible Euler equations [7]; shape-deformation modes are stabilised by the K–V elastic restoring force at the shear modulus scale $G \sim \rho\Gamma^2/r_c^2$ derived in [6]. The stability of the two-zone circulation profile against redistribution of circulation between core and mantle is not addressed analytically here and remains an open problem for future work.

Note on spin quantisation. The emergence of half-integer spin $J = \hbar/2$ requires quantisation of the vortex angular momentum. Within the superfluid analogue the discretisation $\Gamma = nh/m_{\mathrm{p}}$ (Onsager–Feynman, see [6, 7]) discretises the charge; the corresponding quantisation of the angular momentum $L = \rho_0\,\Gamma\,\pi r_0^2$ in units of $\hbar/2$ remains an open problem, deferred to future work.

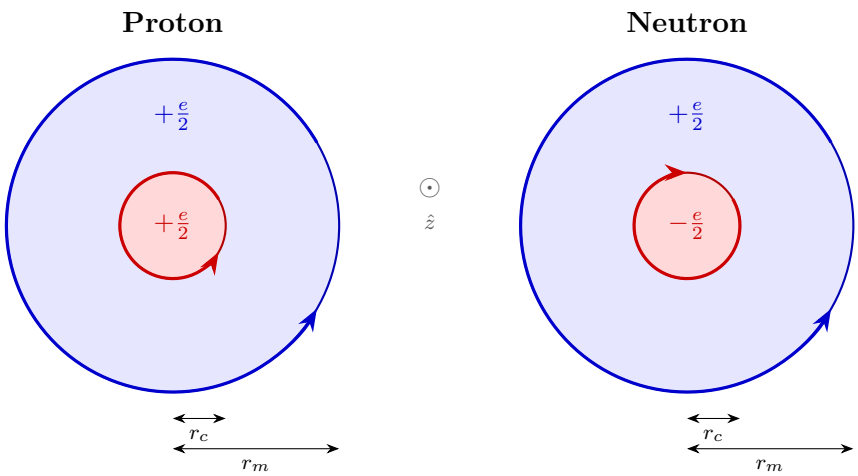

Figure 1: Cross-sectional schematic of the two-zone nucleon model (view along the symmetry axis; not to scale). Red disk: core zone (radius $r_c \approx 0.25\,\mathrm{fm}$); blue annulus: mantle zone (outer radius $r_m \approx 1.3\,\mathrm{fm}$). Both zones belong to a single Hill-type vortex. Curved arrows indicate the rotation direction of each zone. Proton: both zones rotate in the same sense, each carrying charge $+e/2$. Neutron: the core-zone rotation is reversed ($Q_c = -e/2$, $Q_m = +e/2$), giving zero net charge and magnetic moment $\mu_n = -\mu_c + \mu_m = -1.91\,\mu_N$.

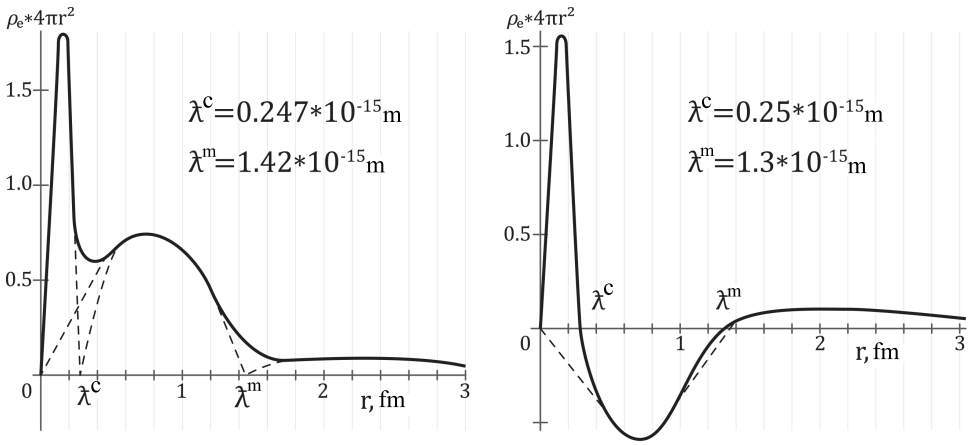

Figure 2: Experimental data on the electric charge distribution in the proton (left) and neutron (right). The vertical axis shows the normalised charge density; the horizontal axis shows the radial distance from the centre.

3 Model Parameters, Consistency, and Hyperon Extensions

3.1 System of Equations for Magnetic Moments

The proton carries total charge $+e$; the neutron carries zero net charge. Within the K–V framework the proton–neutron transition is a reversal of the core-zone rotation only, leaving the mantle zone unchanged. These two constraints together with the single-zone-reversal hypothesis yield the linear system

$$Q_c + Q_m = e \quad \text{(proton)}, \qquad -Q_c + Q_m = 0 \quad \text{(neutron)}, \tag{5}$$

with unique solution $|Q_c| = |Q_m| = e/2$: each zone carries exactly half the elementary charge. Using the K–V charge formula (2), the equal-charge condition reads:

$$|Q_c| = |\rho_0\, r_c\, \Gamma_c| = \tfrac{e}{2}, \qquad |Q_m| = |\rho_0\, r_m\, \Gamma_m| = \tfrac{e}{2}, \tag{6}$$

i.e. $r_c\Gamma_c = r_m\Gamma_m$ (at fixed ρ_0). The signs differ between proton and neutron: in the proton both zones rotate in the same sense ($Q_c > 0$, $Q_m > 0$); in the neutron they are opposite ($Q_c < 0$, $Q_m > 0$), giving net charge zero.

Within the K–V framework this sign assignment acquires a natural geometric interpretation: the isospin quantum number ($I_3 = +\frac{1}{2}$ for the proton, $I_3 = -\frac{1}{2}$ for the neutron) maps directly onto the sign of the core circulation Γ_c, while the mantle circulation $\Gamma_m > 0$ remains unchanged. This identification provides a mechanical picture of isospin within the vortex model.

The additive rules $\mu_p = \mu_c + \mu_m$ (proton, both circulations aligned) and $\mu_n = -\mu_c + \mu_m$ (neutron, core reversed), combined with the experimental values, give the system:

$$\begin{cases} \mu_c + \mu_m = 2.79\,\mu_N, \\ -\mu_c + \mu_m = -1.91\,\mu_N, \end{cases} \tag{7}$$

The unique solution is $\mu_c = 2.35\,\mu_N$ and $\mu_m = 0.44\,\mu_N$. Table 1 summarises the zone parameters.

Table 1: Calculated parameters of the two zones within the model.

Component	Charge (e)	μ/μ_N	Radius (fm)
Core	+0.5 (p), −0.5 (n)	2.35	0.25
Mantle	+0.5	0.44	1.3

The neutron magnetic moment evaluates to

$$\mu_n = -\mu_c + \mu_m = -2.35 + 0.44 = -1.91\,\mu_N, \tag{8}$$

reproducing the experimental value including sign.

3.2 Consistency Check

The selection criterion is agreement with the experimental ratio of radii $r_m/r_c \approx 5.2$ [4, 5]. From the magnetic moment formula (3), $\mu \sim \rho_0\, V\, \omega^2 r_0^2 \sim \rho_0\, r_0^5\, \omega^2$. Using $\Gamma = 2\pi r_0^2 \omega$ to eliminate ω, this becomes

$$\mu \;\propto\; \frac{\rho_0\, \Gamma^2\, r_0}{4\pi^2}. \tag{9}$$

Therefore

$$\frac{\mu_c}{\mu_m} = \frac{\Gamma_c^2\, r_c}{\Gamma_m^2\, r_m}. \tag{10}$$

Applying the equal-charge condition $r_c\Gamma_c = r_m\Gamma_m$, i.e. $\Gamma_c/\Gamma_m = r_m/r_c$:

$$\frac{\mu_c}{\mu_m} = \left(\frac{r_m}{r_c}\right)^2 \cdot \frac{r_c}{r_m} = \frac{r_m}{r_c}. \tag{11}$$

The model therefore predicts that the ratio of magnetic moments equals the ratio of vortex radii: $\mu_c/\mu_m = r_m/r_c \approx 5.2$.

Note on the range of validity of eq. (11). Equation (10) involves Γ_c^2 and Γ_m^2 (squares), so the derivation is insensitive to the relative signs of the circulations. Consequently, eq. (11) holds as a magnitude relation $|\mu_c|/|\mu_m| = r_m/r_c$ regardless of whether the two zones co-rotate or counter-rotate. This extends its applicability to the Λ and Ξ sectors (Secs. 3.3–3.5), where the equal-charge condition for a neutral baryon forces opposite-sign circulations.

The predicted value $\mu_c/\mu_m = 2.35/0.44 \approx 5.34$ is in agreement with the geometric ratio $r_m/r_c = 5.2 \pm 1.2$ (see below), confirming the extension of the K–V framework to magnetic moments.

Note on uncertainties. The magnetic moment ratio $\mu_c/\mu_m = 5.34$ is obtained from $\mu_p = 2.7928\,\mu_N$ and $\mu_n = -1.9130\,\mu_N$, which are known to better than $10^{-4}\,\mu_N$; its uncertainty is negligible. The geometric ratio r_m/r_c, by contrast, is extracted from the phenomenological two-component fit to Hofstadter-era electron-scattering data [4, 5]; the characteristic scales $r_c \approx 0.25\,\text{fm}$ and $r_m \approx 1.3\,\text{fm}$ carry uncertainties of roughly $\delta r_c \approx 0.05\,\text{fm}$ and $\delta r_m \approx 0.15\,\text{fm}$ (estimated from the scatter across analyses). Error propagation gives

$$\delta\left(\frac{r_m}{r_c}\right) \approx \frac{r_m}{r_c}\sqrt{\left(\frac{\delta r_c}{r_c}\right)^2 + \left(\frac{\delta r_m}{r_m}\right)^2} \approx 5.2 \times \sqrt{0.04 + 0.013} \approx 1.2, \tag{12}$$

so that $r_m/r_c = 5.2 \pm 1.2$. The predicted value 5.34 lies within this range, confirming consistency, but the comparison is limited by the geometric uncertainty rather than by the precision of the magnetic moment data.

The counterintuitive result $\mu_c > \mu_m$ despite $r_c < r_m$ has a transparent physical interpretation. The equal-charge condition $r_c\Gamma_c = r_m\Gamma_m$ forces the smaller vortex to rotate faster: $\Gamma_c/\Gamma_m = r_m/r_c \approx 5.2$. Since $\mu \propto \Gamma^2 r_0$, the gain from elevated circulation outweighs the penalty of reduced size, giving $\mu_c \propto \Gamma_c^2 r_c \propto r_m^2/r_c > \mu_m \propto \Gamma_m^2 r_m$. The mechanism is analogous to a figure skater pulling their arms inward: reducing the vortex radius at fixed charge (the hydrodynamic angular impulse $r_0\Gamma$) accelerates the circulation and disproportionately amplifies the magnetic moment.

3.3 Testable Prediction: the Λ Hyperon

The prediction $\mu_c/\mu_m = r_m/r_c$ (eq. (11)) acquires new content when applied to the Λ hyperon ($J^P = \frac{1}{2}^+$, $S = -1$, $\mu_\Lambda = -0.613\,\mu_N$ [9]). Because the Λ is an *isospin singlet* ($I = 0$) with no charged partner, the total magnetic moment provides only a single constraint on the two zones. Since $Q_\Lambda = 0$, the equal-charge condition requires opposite-sign zone charges: $Q_c = -Q_m$ with $|Q_c| = |Q_m|$. There is a twofold ambiguity in the sign assignment:

(A) $Q_c = -e/2$, $Q_m = +e/2$ (core negative, mantle positive): $\mu_\Lambda = -\mu_c(\Lambda) + \mu_m(\Lambda)$;

(B) $Q_c = +e/2$, $Q_m = -e/2$ (core positive, mantle negative): $\mu_\Lambda = \mu_c(\Lambda) - \mu_m(\Lambda)$.

The observed $\mu_\Lambda < 0$, combined with $\mu_c > \mu_m$ (which follows from $r_m > r_c$ via eq. (11)), selects assignment (A) uniquely: only $-\mu_c + \mu_m < 0$, whereas (B) gives $\mu_c - \mu_m > 0$, contradicting experiment. This yields:

$$\mu_\Lambda \;=\; -\mu_c(\Lambda) + \mu_m(\Lambda) \;=\; -0.613\,\mu_N. \tag{13}$$

The system is under-determined: $\mu_c(\Lambda)$ and $\mu_m(\Lambda)$ cannot be fixed from this equation alone. Rather than closing it with an *ad hoc* assumption, we recognise this under-determination as a direct statement about what must be measured.

The model predicts that two independently accessible quantities must be equal:

$$\frac{\mu_c(\Lambda)}{\mu_m(\Lambda)} \;=\; \frac{r_m(\Lambda)}{r_c(\Lambda)}. \tag{14}$$

The left-hand side is accessible from the spatial profile of the Λ *magnetisation distribution*, encoded in the magnetic Sachs form factor $G_M^{(\Lambda)}(Q^2)$ at large Q^2; the right-hand side is accessible from the *charge distribution*, encoded in $G_E^{(\Lambda)}(Q^2)$. These two form factors are measured independently in the same $e\Lambda$ scattering experiment. The Λ lifetime ($c\tau \approx 7.89\,\mathrm{cm}$) is sufficient for hyperon-beam or secondary-target configurations, making such measurements experimentally accessible at future facilities such as PANDA at FAIR—in contrast to the Ξ baryons, for which the lifetime and production cross-section present additional experimental challenges (Sec. 3.5). Equation (14) combined with eq. (13) then uniquely determines the component moments:

$$\mu_m(\Lambda) = \frac{0.613}{x-1}, \qquad \mu_c(\Lambda) = \frac{0.613\,x}{x-1}, \qquad x \equiv \frac{r_m(\Lambda)}{r_c(\Lambda)}. \tag{15}$$

Once x is measured, both component moments are predicted with no free parameters. The equality (14) is the non-trivial content of the model: it can fail, and its failure would falsify the equal-charge two-zone ansatz for the Λ.

As a numerical illustration, Table 2 shows the predicted component moments for two hypothetical values of x. If the Λ vortex geometry is similar to the nucleon ($x \approx 5.2$), then $\mu_m(\Lambda) \approx 0.145\,\mu_N$ and $\mu_c(\Lambda) \approx 0.758\,\mu_N$; if the structure is more compact ($x \approx 3$), then $\mu_m(\Lambda) \approx 0.307\,\mu_N$ and $\mu_c(\Lambda) \approx 0.920\,\mu_N$. These two scenarios predict qualitatively different spatial profiles of the magnetisation density and are distinguishable in principle from the Q^2-dependence of $G_M^{(\Lambda)}(Q^2)$.

Table 2: Predicted Λ-hyperon vortex component moments as a function of the geometric ratio $x = r_m(\Lambda)/r_c(\Lambda)$, using $\mu_\Lambda = -0.613\,\mu_N$ and eqs. (13)–(15).

$x = r_m/r_c$	$\mu_c(\Lambda)/\mu_N$	$\mu_m(\Lambda)/\mu_N$
3.0	0.920	0.307
4.0	0.817	0.204
5.2 (nucleon-like)	0.758	0.145
7.0	0.716	0.102

3.4 Extension to the Σ Isospin Triplet

The Σ baryons ($I = 1$, $J^P = \frac{1}{2}^+$) form an isospin triplet: Σ^+ (charge $+e$), Σ^0 (charge 0), and Σ^- (charge $-e$). Their precisely measured magnetic moments [9] are $\mu_{\Sigma^+} = +2.458\,\mu_N$ and $\mu_{\Sigma^-} = -1.160\,\mu_N$.

Unlike the nucleon isospin doublet ($\Delta I_3 = 1$ step), the Σ triplet spans $\Delta I_3 = 2$ from Σ^+ to Σ^-. In the K–V ladder picture, each unit step in I_3 reverses one component: the first step ($\Sigma^+ \to \Sigma^0$, core reversed) gives charges $(-e/2, +e/2)$ and net zero; the second step ($\Sigma^0 \to \Sigma^-$, mantle reversed) gives charges $(-e/2, -e/2)$ and net $-e$. Applying this double-reversal scheme directly to the Σ^+/Σ^- pair:

$$\mu_{\Sigma^+} = \mu_c^{(\Sigma)} + \mu_m^{(\Sigma)} = 2.458\,\mu_N, \qquad \mu_{\Sigma^-} = -\mu_c^{(\Sigma)} - \mu_m^{(\Sigma)} = -1.160\,\mu_N. \tag{16}$$

These two equations are inconsistent ($2.458 \neq 1.160$), indicating that the equal-charge two-zone ansatz of the nucleon sector does *not* transfer directly to the Σ triplet. This is physically expected: the Σ contains an s-quark, and the different quark content changes both the charge distribution geometry and the internal circulation strengths.

The inconsistency is itself a meaningful result: it implies that the charge distribution radii $r_c^{(\Sigma)}$ and $r_m^{(\Sigma)}$ are *not* simply related to those of the nucleon, and that their independent measurement (from $e\Sigma$ scattering form factors, accessible at future facilities such as JLab, HIAF, or PANDA at FAIR) would be required to close the system.

The degree of inconsistency can be quantified by a sum rule: in the symmetric double-reversal scheme the sum $\mu_{\Sigma^+} + \mu_{\Sigma^-}$ should vanish identically, whereas experiment gives

$$\mu_{\Sigma^+} + \mu_{\Sigma^-} = 2.458 - 1.160 = +1.298\,\mu_N \neq 0. \tag{17}$$

This non-zero residual is a direct measure of the s-quark's effect on the reversal symmetry of the nucleon sector: it tells us by how much the circulation balance between core and mantle differs between Σ^+ (quark content uus) and Σ^- (dds). In K–V terms it implies that the charge distributions of the two members are not mirror images of each other. A direct consequence is that the electric form factors $G_E^{(\Sigma^+)}(Q^2)$ and $G_E^{(\Sigma^-)}(Q^2)$ should differ not only in sign but in *shape*—a prediction that is falsifiable at future hyperon scattering facilities independently of any assumption about r_c and r_m.

Once $r_m^{(\Sigma)}/r_c^{(\Sigma)}$ is measured for each charge state *separately* (since the Σ^+ and Σ^- charge distributions differ in shape, each has its own geometric ratio), the model predicts the key relation for each member individually:

$$\frac{\mu_c^{(\Sigma^\pm)}}{\mu_m^{(\Sigma^\pm)}} = \frac{r_m^{(\Sigma^\pm)}}{r_c^{(\Sigma^\pm)}}, \tag{18}$$

which provides a non-trivial consistency check of the K–V framework beyond the nucleon sector.

3.5 Extension to the Ξ Cascade

The Ξ baryons ($I = \frac{1}{2}$, $J^P = \frac{1}{2}^+$, strangeness $S = -2$) form a doublet with measured moments [9] $\mu_{\Xi^0} = -1.250\,\mu_N$ and $\mu_{\Xi^-} = -0.6507\,\mu_N$.

The Ξ doublet provides a stronger constraint on the framework than the Λ singlet: two independent magnetic moment measurements fully determine the system, leaving no free parameters, whereas for Λ a single measurement left the system underdetermined and

the model could only predict a relation between observables. This makes the Ξ sector a more decisive test—and, as shown below, a decisive *failure*.

Unlike the nucleon, for which the two-zone charge distribution is directly established by electron-scattering data [4, 5], no analogous measurement exists for the Ξ baryons: their short lifetime ($\sim 10^{-10}$ s) precludes their use as fixed targets, and the electromagnetic form factors $G_E^{(\Xi)}(Q^2)$ and $G_M^{(\Xi)}(Q^2)$ have not been measured. The two-zone structure and the equal-charge condition are therefore not empirically motivated for the Ξ sector; they constitute an untested assumption when applied here by analogy.

Applying the equal-charge ansatz $|Q_c| = |Q_m|$ with the constraint $Q_c^{(\Xi)} + Q_m^{(\Xi)} = 0$ (charge of Ξ^0) gives $Q_c^{(\Xi)} = +e/2$ and $Q_m^{(\Xi)} = -e/2$: the two zones carry *opposite-sign* charges. Unlike the nucleon, where both zones co-rotate ($\Gamma_c > 0$, $\Gamma_m > 0$), opposite-sign charges imply opposite-sign circulations ($\Gamma_c > 0$, $\Gamma_m < 0$). Under the K–V sign convention (eq. (4)), this gives $\mu_c^{(\Xi)} > 0$ and $\mu_m^{(\Xi)} < 0$.

Applying the core-reversal hypothesis (Ξ^0 as the $I_3 = +\frac{1}{2}$ reference state, Ξ^- obtained by core reversal) with these sign constraints:

$$|\mu_c^{(\Xi)}| - |\mu_m^{(\Xi)}| = -1.250\,\mu_N \quad (\Xi^0), \qquad -|\mu_c^{(\Xi)}| - |\mu_m^{(\Xi)}| = -0.6507\,\mu_N \quad (\Xi^-). \tag{19}$$

Adding these equations gives $-2|\mu_m^{(\Xi)}| = -1.901\,\mu_N$, hence $|\mu_m^{(\Xi)}| = 0.950\,\mu_N$; subtracting gives $2|\mu_c^{(\Xi)}| = -0.599\,\mu_N < 0$, which has no physical solution.

The Ξ magnetic moment data are therefore *incompatible* with the equal-charge two-zone ansatz under the K–V sign convention, analogously to the Σ triplet (Sec. 3.4). The inconsistency is more fundamental than for Σ: there it arose from the triplet structure, whereas here the equal-charge condition applied to the neutral member Ξ^0 itself forces opposite-sign zone circulations that cannot simultaneously accommodate both measured moments. Whether a modified two-zone ansatz—with unequal zone charges or a different reversal scheme—could fit the Ξ data remains an open question. In K–V terms, two structural hypotheses naturally present themselves. First, the two-zone model may be insufficient: each strange quark could require an additional circulation scale, so that a three-zone structure with radii $r_c < r_{\rm int} < r_m$ would appear as three distinct form factor scales in $G_E^{(\Xi)}(Q^2)$. Second, the vortex topology itself may change for $S = -2$: instead of a single Hill vortex, double strangeness may be associated with two linked circulation loops, altering the reversal mechanism at a qualitative level. Both scenarios predict that the Ξ charge form factor will differ *qualitatively*—not merely quantitatively—from the nucleon, which is directly testable via the Q^2-shape of $G_E^{(\Xi)}(Q^2)$ at future facilities such as PANDA at FAIR.

3.6 Strangeness as a Structural Probe: A Research Programme

The results of Secs. 3.3–3.5 form a coherent pattern that can be organised into a systematic research programme. The equal-charge two-zone ansatz succeeds exactly for $S = 0$, survives (in a single-equation, underdetermined form) for the Λ ($S = -1$, $I = 0$), fails quantitatively for the Σ ($S = -1$, $I = 1$), and fails fundamentally for the Ξ ($S = -2$). This hierarchy suggests that each unit of strangeness introduces a structural complication that the simplest two-zone equal-charge model cannot accommodate, and that the *degree* of failure itself carries physical information about the internal structure of strange baryons.

Table 3 summarises the status of the equal-charge ansatz across the baryon octet, together with the structural prediction that follows from each outcome.

Table 3: Status of the equal-charge two-zone ansatz across the baryon octet, organised by strangeness. The sum-rule deviation Δ is defined as the value that should vanish in a symmetric reversal scheme. A dash (—) indicates that no reversal partner exists.

S	Baryon	Ansatz status	Δ	Structural prediction
0	p, n	exact	0 (by construction)	Two zones, $r_m/r_c \approx 5.2$
-1	Λ	underdetermined	—	$\mu_c/\mu_m = r_m/r_c$ (eq. (14))
-1	$\Sigma^{\pm}$	fails	$+1.298\,\mu_N$	$G_E^{(\Sigma^+)} \neq G_E^{(\Sigma^-)}$ in shape
-2	$\Xi^{0,-}$	impossible	no solution	≥ 3 zones or different topology

The sum-rule deviation $\Delta = \mu_{\Sigma^+} + \mu_{\Sigma^-} = +1.298\,\mu_N$ (eq. (17)) is a quantitative diagnostic: it measures the asymmetry between the charge distributions of Σ^+ (uus) and Σ^- (dds) that the s-quark introduces. A systematic measurement of the analogous sum rule $\mu_B + \mu_{\bar{B}}$ across the baryon octet would map the strangeness dependence of vortex-structure asymmetry within the K–V picture.

For the Ξ doublet, the absence of any physical solution is a stronger statement: not a quantitative mismatch but a qualitative change of regime. This motivates two specific experimental targets. First, determine whether $G_E^{(\Xi)}(Q^2)$ exhibits two or more distinct form-factor scales (two-zone vs. multi-zone). Second, compare the Q^2-slopes of $G_E^{(\Xi^0)}$ and $G_E^{(\Xi^-)}$ to test whether their charge distributions are mirror images or differ in shape—a test that is independent of any model assumption about the number of zones.

In this way, the regions where the K–V model *fails* function as *diagnostic tools*: each inconsistency translates into a concrete, falsifiable prediction about the charge-distribution geometry of the corresponding baryon, accessible at future hyperon scattering facilities.

4 Discussion of Results and Educational Value

The presented model offers a unified and intuitively understandable description of the key experimental facts: charge distribution, anomalous magnetic moments, and baryon sizes. The K–V mechanical analogy of [6] provides a firm theoretical footing: the charge formula $Q = \rho_0 r_0 \Gamma$ is not postulated but derived from the constitutive law of the viscoelastic medium and Kelvin's circulation theorem. The magnetic moment formula $\mu = \rho_0 V(\omega r_0)^2$ is a natural extension, expressing μ through the same microscopic parameters (ρ_0, r_0, Γ) that define the charge. Table 4 compares the model predictions against experiment and two established frameworks for the nucleon sector.

4.1 Comparison with Other Phenomenological Models

Several established models describe nucleon magnetic moments, each with distinct strengths. The **non-relativistic quark model** (NQM) reproduces μ_p and μ_n via constituent quark spins ($\mu_p = \frac{4}{3}\mu_u - \frac{1}{3}\mu_d$, $\mu_n = \frac{4}{3}\mu_d - \frac{1}{3}\mu_u$), but takes the quark magnetic moments as free parameters and offers no geometrical connection between the charge distribution and the magnetic moment ratio. The **MIT bag model** treats quarks as free fields inside a confining sphere of radius $R \sim 1\,\text{fm}$; it predicts anomalous moments approximately but requires separate treatment of the confinement scale and the magnetic properties, and does not exhibit a natural two-scale internal structure in the way observed in the Hofstadter data. The **Skyrme model** represents the nucleon as a quantised topological

Table 4: Comparison of nucleon parameters in various models.

Parameter	Experiment	Quark Model	Vortex (K–V) Model
μ_p/μ_N	2.79	2.79	2.79
μ_n/μ_N	-1.91	-1.91	-1.91
r_m/r_c	5.2 ± 1.2	n.a.†	5.34
r_c, fm	$\sim$0.25	n.a.†	0.25
r_m, fm	$\sim$1.3	n.a.†	1.3

† The non-relativistic quark model predicts magnetic moments from constituent quark spins and does not define a two-scale radial geometry. The MIT bag model has a single confinement radius $R \approx 1\,\text{fm}$ and does not exhibit a compact core–extended mantle structure. Note that for the K–V model μ_p and μ_n are *inputs* to the system (7), whereas the quark model predicts them from constituent quark moments.

soliton; it correctly predicts the sign of the ratio μ_p/μ_n from topological arguments but has no direct hydrodynamic interpretation of the charge distribution geometry.

The K–V two-zone model occupies a complementary niche. Its distinctive feature is the parameter-free relation $\mu_c/\mu_m = r_m/r_c$ (eq. (11)), which connects two independently measured quantities—the magnetic moment ratio from spin measurements and the geometric ratio from electron scattering—through a single mechanical constraint (the equal-charge condition). None of the three models above yields an analogous geometrical prediction. The price of this simplicity is a limited scope: in its nucleon sector the model relies on the equal-charge condition, whose extension to hyperons requires independent measurement of their charge-distribution geometry (see §3.3–3.5), and the K–V framework itself awaits independent peer review [6].

The use of classical hydrodynamic imagery offers a significant **advantage for educational applications**: the model can be used to create visual aids and simulations, helping students form an initial intuitive understanding of hadron structure before moving on to the complex mathematical apparatus of QCD. It is important to emphasise that the model is **phenomenological and does not claim to be fundamental**, but serves as a convenient tool for describing and systematising experimental data.

5 Development Prospects and Experimental Verification

To test the model and develop it further, the following experiments are proposed:

1. **Measurement of the magnetisation distribution inside the nucleon.** The model predicts different magnetic moment densities in the core and mantle (ratio ≈ 5.34). This translates into a specific two-scale behaviour of the magnetic Sachs form factor $G_M^p(Q^2)$: a slower fall-off at intermediate Q^2 (dominated by the extended mantle, $r_m \sim 1.3\,\text{fm}$) giving way to a harder component at large Q^2 (the compact core, $r_c \sim 0.25\,\text{fm}$, $Q^2 \gtrsim 10\,\text{GeV}^2$). The high-precision recoil-polarisation measurements at Jefferson Lab Hall C [11, 12, 13] have established that the ratio G_E^p/G_M^p falls linearly with Q^2 above $1\,\text{GeV}^2$, in clear disagreement with the simple dipole ansatz. This behaviour is precisely what a two-scale internal structure predicts: the mantle ($r_m \sim 1.3\,\text{fm}$) dominates G_M^p at moderate Q^2, while the compact core ($r_c \sim 0.25\,\text{fm}$) contributes a harder component visible at $Q^2 \gtrsim 10\,\text{GeV}^2$. Re-analysing these data

within the two-component parameterisation $G_M^p(Q^2) = G_M^{(c)}(Q^2) + G_M^{(m)}(Q^2)$ with independently fitted core and mantle form factors offers a direct experimental test of the model's geometric structure.

2. **Search for kinematic features in deep-inelastic scattering.** One might expect that the transition between scattering on the compact core and the extended mantle could manifest itself in sharper changes in cross-sections compared to predictions of models based on parton distributions.

3. **Observation of rotational effects in high-energy collisions.** The vortex nature of nucleons might manifest itself in specific particle correlations in the final state [8]. Strikingly, global Λ hyperon polarization consistent with macroscopic vortical flow has already been measured by the STAR Collaboration in $\sqrt{s_{NN}} = 7.7$–$39\,\mathrm{GeV}$ Au+Au collisions [10]: the observed polarization reaches $\sim 1.08\%$, corresponding to an estimated angular velocity of the quark–gluon plasma of $\omega \sim 10^{22}\,\mathrm{s}^{-1}$, making it the most rapidly rotating fluid known in nature. Within the present framework, this observation demonstrates that macroscopic vortex structures form in nuclear matter under extreme conditions, and motivates a search for connections between the microscopic vortex structure of individual nucleons and the bulk angular momentum of the plasma.

4. **Extension to other hadrons.** Sections 3.3–3.5 develop the model for the Λ, Σ, and Ξ hyperons. For the Λ, the predicted relation $\mu_c/\mu_m = r_m/r_c$ (eq. (14)) offers a concrete falsifiable test accessible via the magnetic Sachs form factor $G_M^{(\Lambda)}(Q^2)$. For both Σ and Ξ, the equal-charge ansatz is found inconsistent with the measured magnetic moments; in the Ξ case this is compounded by the complete absence of charge-distribution data, making a measurement of $G_E^{(\Xi)}(Q^2)$ the necessary first step. Beyond the spin-$\frac{1}{2}$ baryons, testing whether mesons (e.g. ρ, ω) admit an analogous two-zone decomposition would probe the universality of the K–V framework across the hadron spectrum.

6 Conclusion

The Kelvin–Voigt mechanical analogy of [6] has been extended to a self-consistent description of nucleon magnetic moments. In the K–V framework the electric charge is the circulation invariant $Q = \rho_0 r_0 \Gamma$ (eq. (1)), and the magnetic moment is $\mu = \rho_0 V(\omega r_0)^2$ (eq. (3)), with both quantities expressed through the same microscopic vortex parameters. The equal-charge condition $r_c\Gamma_c = r_m\Gamma_m$ yields the key prediction $\mu_c/\mu_m = r_m/r_c$ (eq. (11)). The predicted value 5.34 is consistent with the geometric ratio $r_m/r_c = 5.2 \pm 1.2$ (eq. (12)); the comparison is limited by the uncertainty of the 1960s charge-distribution data rather than by the precision of the magnetic moment measurements.

The same geometric relation $\mu_c/\mu_m = r_m/r_c$ has been extended to the hyperon sector. For the Λ (isospin singlet), it provides a falsifiable prediction (eq. (14)) that connects the Λ magnetic and charge form factors and can be tested at future electron–hyperon scattering facilities. For the Σ isospin triplet, the double-reversal scheme applied to Σ^+/Σ^- is found to be inconsistent with experiment, indicating that the equal-charge ansatz does not transfer directly from the nucleon sector, and that an independent measurement of the Σ charge distribution is required. For the Ξ doublet, the equal-charge ansatz applied to the neutral member Ξ^0 forces opposite-sign zone circulations, leading to a system with

no physical solution for the zone magnetic moments—an inconsistency analogous to, and more fundamental than, the Σ case. No charge-distribution data exist for the Ξ, making a measurement of $G_E^{(\Xi)}(Q^2)$ at future facilities the necessary prerequisite for any K–V prediction in the double-strangeness sector.

The pattern of outcomes across the baryon octet—exact agreement at $S = 0$, underdetermination at $S = -1$ (Λ), quantitative failure at $S = -1$ (Σ), and fundamental impossibility at $S = -2$ (Ξ)—constitutes in itself a systematic research programme (Sec. 3.6, Table 3): each inconsistency translates into a concrete structural prediction about the corresponding baryon, falsifiable through independent measurements of the electromagnetic form factors at future hyperon scattering facilities.

The constructed phenomenological two-zone model offers an intuitive picture of baryon internal structure and significant advantages for educational applications. Explicit proposals for experimental verification are formulated.

References

[1] Hofstadter, R. (1956). Electron Scattering and Nuclear Structure. *Reviews of Modern Physics*, 28(3), 214–254.

[2] Ernst, D. J., et al. (1994). Nucleon form factors from electron-nucleon scattering. *Physical Review C*, 50(2), 910–922.

[3] Schopper, H. (1961). Die elektromagnetische Struktur des Protons und Neutrons. *Physikalische Blätter*, 21(10), 454–462.

[4] Hofstadter, R., & Herman, R. (1961). Electric and magnetic structure of the proton and neutron. *Physical Review Letters*, 6(6), 293–296.

[5] Hand, L. N., Miller, D. G., & Wilson, R. (1963). Electric and magnetic form factors of the nucleon. *Reviews of Modern Physics*, 35(2), 335–349.

[6] Losinets, D. S. (2026). Maxwell's Equations from Vortex Mechanics: A Kelvin–Voigt Analogy. *Zenodo*. https://doi.org/10.5281/zenodo.19135635

[7] Donnelly, R. J. (1991). *Quantized Vortices in Helium II*. Cambridge University Press.

[8] Liang, Z.-T., & Wang, X.-N. (2005). Globally polarized quark-gluon plasma in non-central A+A collisions. *Physical Review Letters*, 94(10), 102301.

[9] Workman, R. L., et al. (Particle Data Group) (2022). Review of Particle Physics. *Progress of Theoretical and Experimental Physics*, 2022, 083C01. https://doi.org/10.1093/ptep/ptac097

[10] STAR Collaboration: Adamczyk, L., et al. (2017). Global Λ hyperon polarization in nuclear collisions. *Nature*, 548, 62–65. https://doi.org/10.1038/nature23004

[11] Jones, M. K., et al. (2000). G_{Ep}/G_{Mp} ratio by polarization transfer in $\vec{e}\,p \to e\,\vec{p}$. *Physical Review Letters*, 84(7), 1398–1402. https://doi.org/10.1103/PhysRevLett.84.1398

[12] Gayou, O., et al. (2002). Measurement of G_{Ep}/G_{Mp} in $\vec{e}\,p \to e\,\vec{p}$ to $Q^2 = 5.6\,\mathrm{GeV}^2$. *Physical Review Letters*, 88, 092301. https://doi.org/10.1103/PhysRevLett.88.092301

[13] Puckett, A. J. R., et al. (2010). Recoil polarization measurements of the proton electromagnetic form factor ratio to $Q^2 = 8.5\,\mathrm{GeV}^2$. *Physical Review Letters*, 104, 242301. https://doi.org/10.1103/PhysRevLett.104.242301